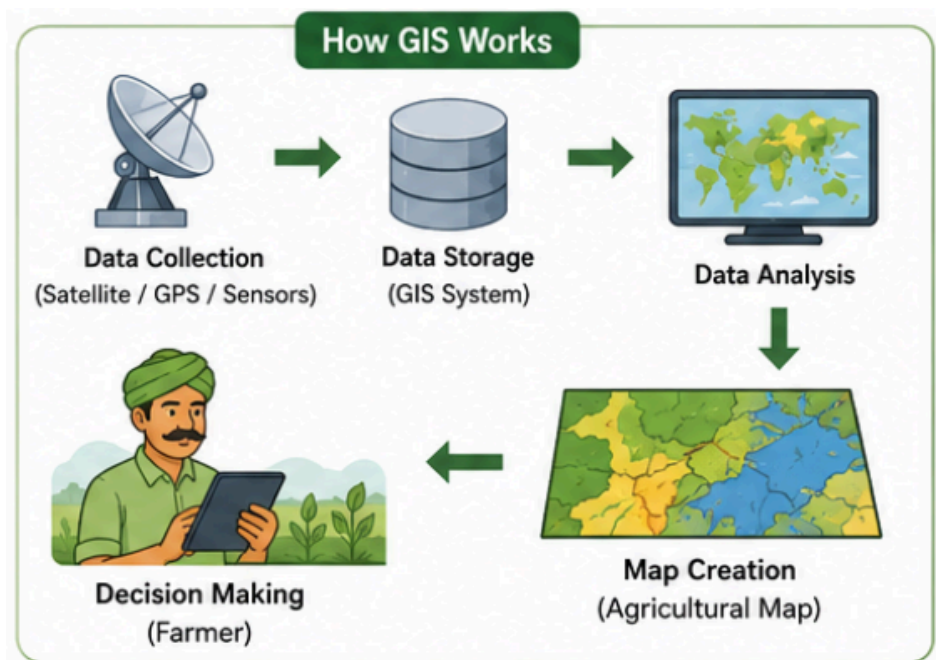


GIS (Geographic Information System)

- GIS stands for Geographic Information System.
- It is a computer-based system used to **collect, store, manage, analyze, and display geographical information**.
- In agriculture, GIS helps farmers and agricultural departments to study **soil, crops, water, weather, and land** using maps and data.



1. Crop Monitoring

GIS helps to monitor the **location and growth of crops**.

Uses:

- Shows crop areas on maps.
 - Monitors crop growth.
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2. Soil Analysis and Mapping

GIS is used to create **soil maps**.

It can show:

- Soil type
 - Soil moisture
 - Soil fertility
 - Nutrient levels
 - Soil salinity
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3. Water Management

GIS helps in managing **water resources and irrigation**.

Uses:

- Maps rivers, ponds, wells, and canals.
 - Identifies areas that need irrigation.
 - Helps plan irrigation systems.
 - Helps reduce water wastage.
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4. Weather Analysis

GIS can combine weather data with maps.

It can display:

- Temperature
 - Rainfall
 - Humidity
 - Wind conditions
 - Drought area
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5. Precision Agriculture

GIS is an important tool for **precision agriculture and improving crop production**.

It combines information from:

- GPS
- Remote sensing
- Ground sensors
- Satellite images

Advantages of GIS in Agriculture

- Helps in **better decision-making**.
- Saves **time and resources**.
- Helps manage large areas.
- Reduces water and fertilizer wastage.